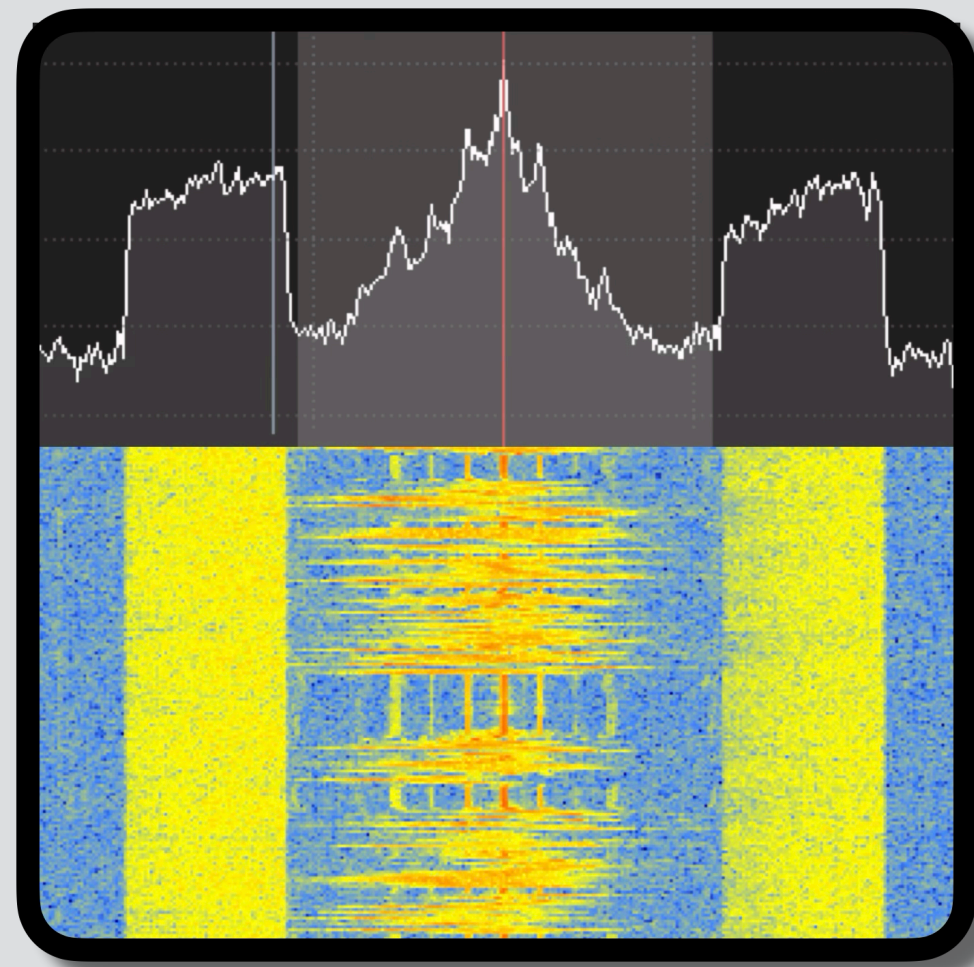
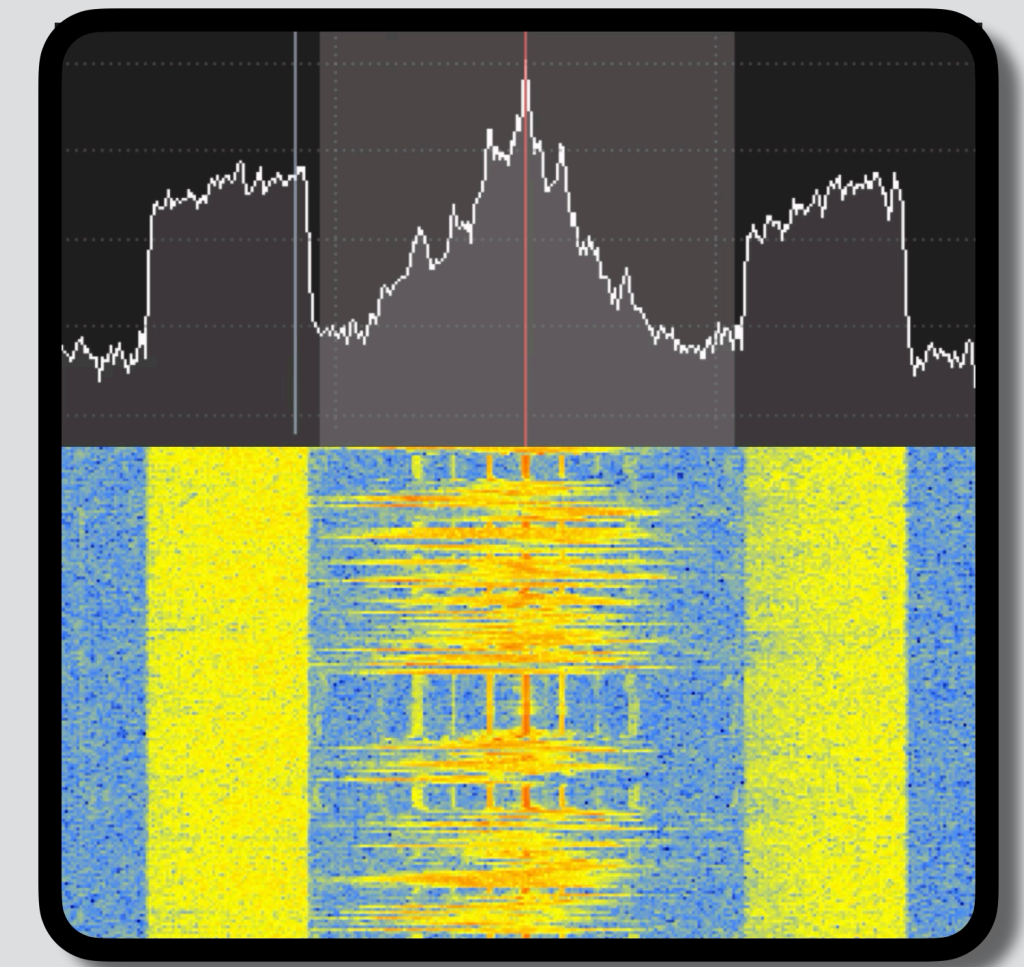




# EE121

*& Analog*



## Introduction to Digital Communication Systems

### Lecture 11B

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# Today

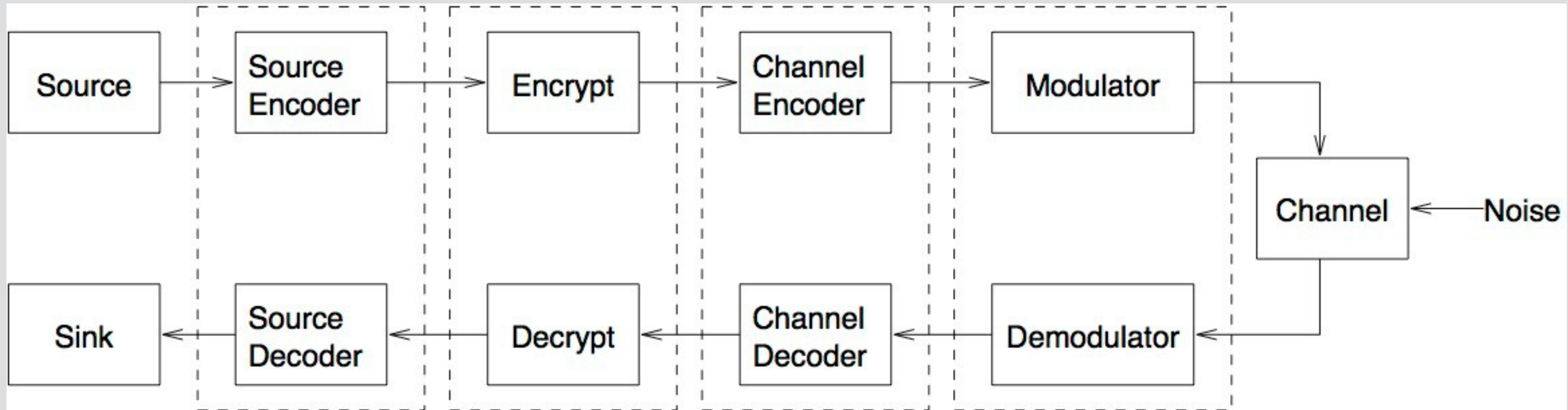
- Last time:
  - Distance Rule (finish)
  - Receiver Realization
  - Soft Decoding
  - Lab 8
- Today: Andrews notes (Bcourses)
  - Error Detection and Correction
  - Parity and CRC
  - Coding

# Error Protection: Detection and Correction

- Communications channels are subject to noise
- Noise can cause digital signals to be received as different values
  - Bits can be flipped
  - Points in the constellation can shift
- Given errors:
  - We would like to detect or correct
  - Add redundancy to do that



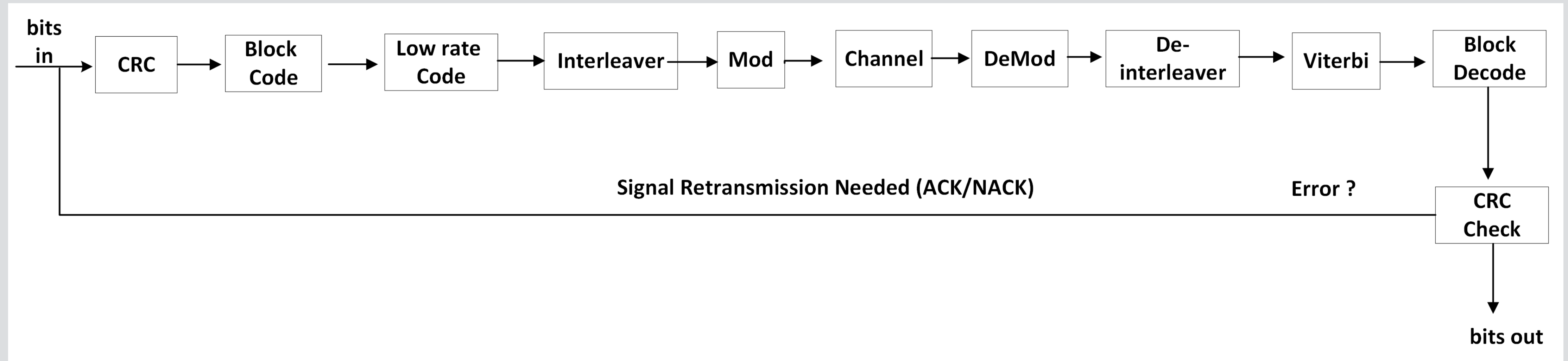
# Comm systems



- So far we focused on the modulator/demodulators
- Error correction involves channel encoding/decoding



# Title Text



- Several Layers of error detection and correction
  - CRC for detection
  - Block codes - Reed Solomon, Hamming
  - Low rate inner codes - Convolutional, LDPC

# Error Control Classification

- Type of errors: random, burst, catastrophic
- Type of error control: Detection, Correction
- Type of Codes: Block, convolutional
- Type of decoder: Algebraic, probabilistic

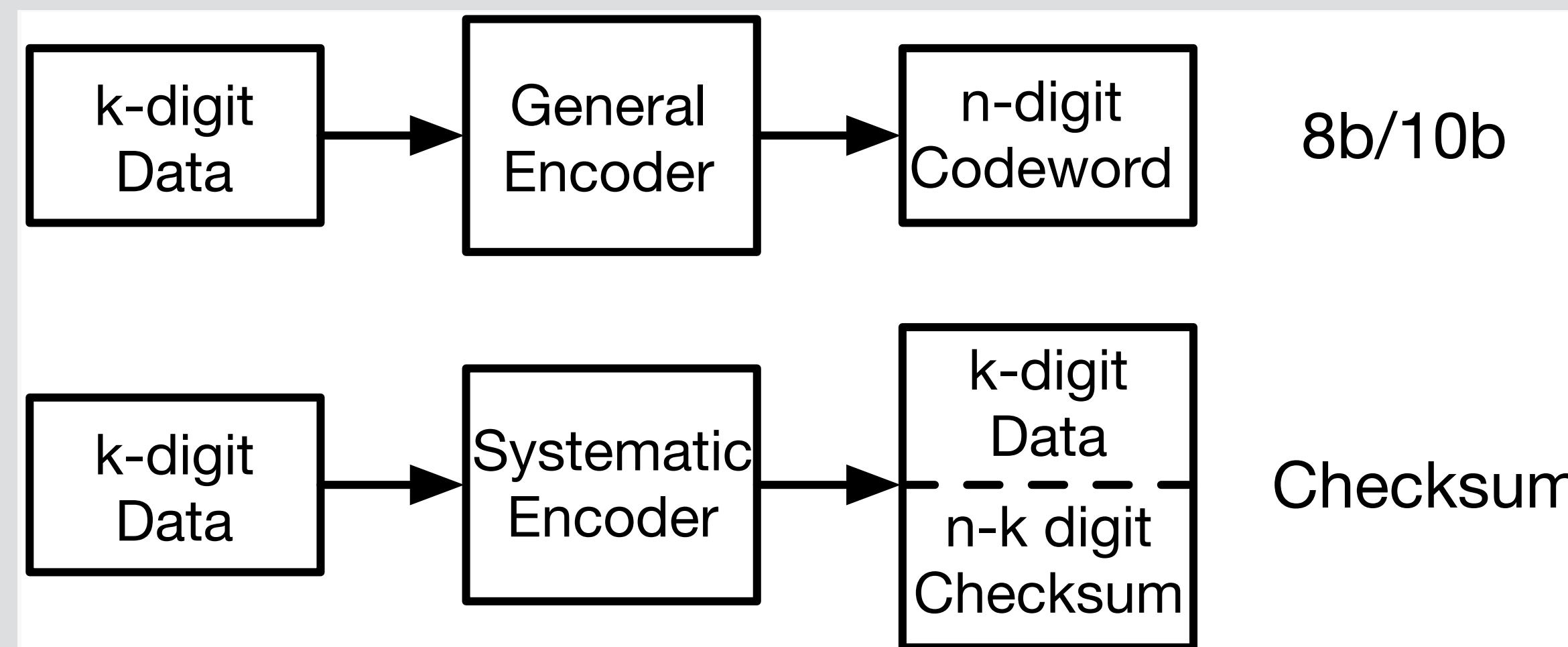
# Types of Error Protection

- Error Detection:
  - Goal: avoid accepting faulty data!
  - Lost data is unfortunate, wrong data can be disastrous
  - Solution: checksum included (packets, frames, sectors)
    - If any part of the message is altered, then the checksum is not valid (with high probability)
- Forward error correction (FEC or ECC)
  - Adds redundancy to the message, send more bits!
  - Receiver estimates the most likely message, even with errors



# Block Codes

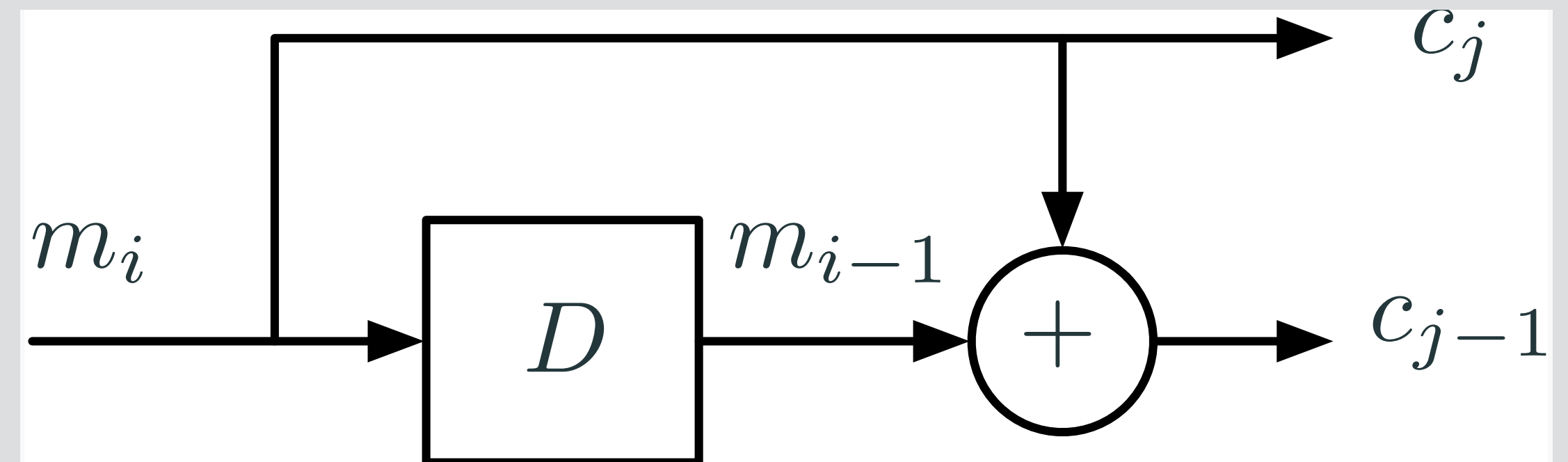
- $k$ -digit blocks of information are encoded into  $n$ -digit codewords  $n \geq k$  by adding  $p = n - k$  check digits
- Blocks are independent from each other
- An encoding where digits are unchanged is *systematic*





# Convolutional Codes

- A “filter” adds additional bits to the bitstream
- Each  $n$ -bit codeword block depends on the current codeword and on previous  $m$  codewords.
  - $m$  is the memory order
  - $r$  is rate,  $1/r$  redundancy



Rate 1/2, with  $m = 1, n = 2$

- Example: two outputs per input bit

# Parity Check

- Used to verify the integrity of a block
- Append a check bit to the data so codeword have the same parity — even or odd

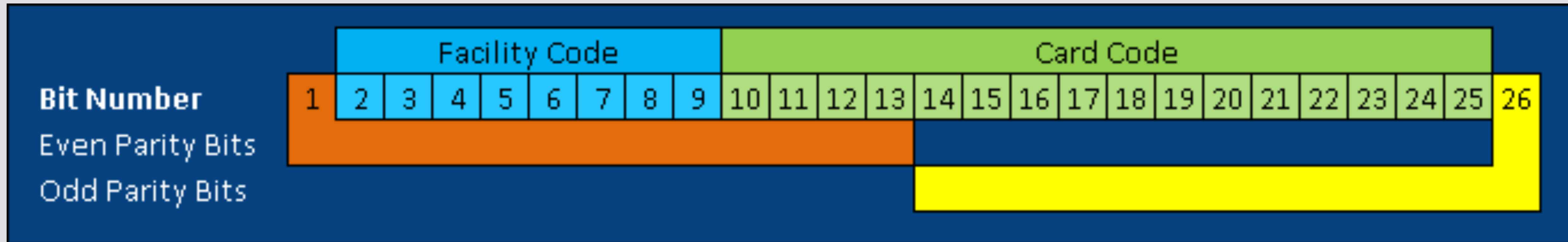
$$c_1 \oplus c_2 \oplus \cdots \oplus c_N \oplus c_p = 0 \quad \Rightarrow \quad c_1 \oplus c_2 \oplus \cdots \oplus c_N = c_p$$

$$c_1 \oplus c_2 \oplus \cdots \oplus c_N \oplus c_p = 1 \quad \Rightarrow \quad c_1 \oplus c_2 \oplus \cdots \oplus c_N \oplus 1 = c_p$$

- Any single bit (or odd number ) error can be detected

# Example: HID RFID

- HID Corporate 1000 cards use
  - 2, odd and even parity bits for 26bit card
  - 3, even and 2 odd parity bits for 35bit card



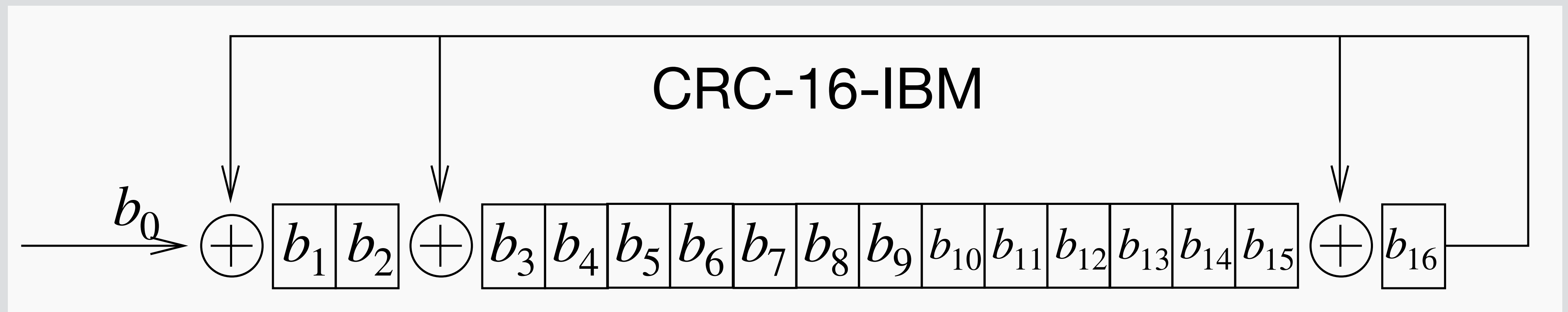
$$c_1 = c_2 \oplus c_3 \oplus \dots \oplus c_{13}$$

$$c_{26} = c_{10} \oplus c_{11} \oplus \dots \oplus c_{25} \oplus 1$$



# Cyclic Redundancy Check (CRC)

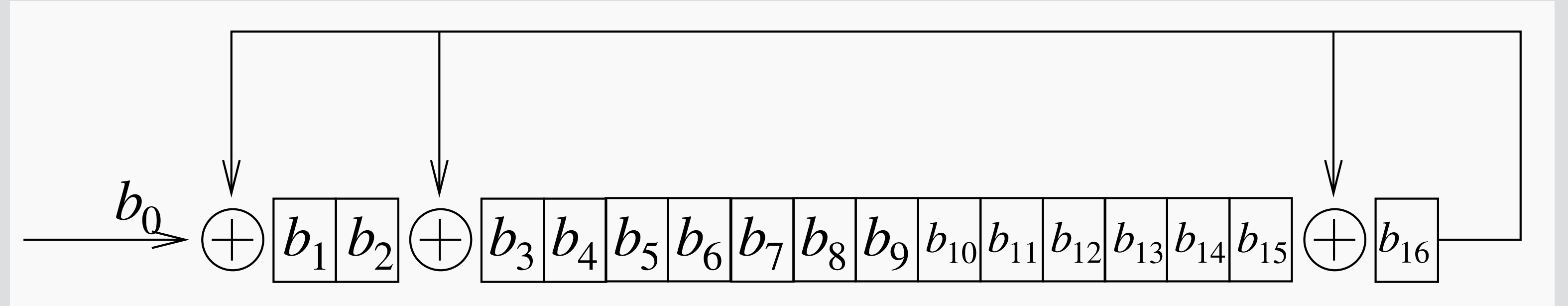
- CRC generators are implemented as a linear-feedback shift registers with XOR operations at various locations



$$g(D) = 1 + D^2 + D^{15} + D^{16}$$

- Polynomial: Every delay is power of 2 - some powers are XORed

# CRC Error Detection



- At TX:
  - Whole packet of bits is input, in serial to the generator
  - The state at the end of the packet is:  $b = (b_1, b_2, \dots, b_{16})$
  - Transmit 16 CRC bits at the end of the packet as a designated part (link layer)
- At RX:
  - Decode the packet
  - Run the decoded bits through the  $g(D)$ , the CRC generator
  - Compare against CRC part of the packet.



# CRC

- Lots of different polynomials are possible. Common:
  - CRC-16-IBM:  $D^{16} + D^{15} + D^2 + 1$
  - CRC-CCITT:  $D^{16} + D^{12} + D^5 + 1$
- Can be applied to any length block  $< 2^N - 1$  bits
- Used in Bluetooth, GSM, X.25, ACARS

# CRC Analysis

- Miss Detection - there are errors, but CRC checks out :-(

For CRC-K: assume iid Bernoulli (ballpark)

$$\begin{aligned} P[\text{miss}] &\approx 2^{-K} \\ &\approx 2^{-16} = 5 \times 10^{-5} \end{aligned}$$

For ethernet, CRC length is 32  $\Rightarrow P[\text{miss}] \approx 2^{-32} = 2.3 \times 10^{-10}$

If an error is missed, then packet is assumed correct. Not cool for storing or sending files!!!!

Sometimes there's also CRC at the application level

# CRC Analysis

- False Alarm - No packet errors, but CRC bits have errors

For CRC-K, with  $P_e = 10^{-5}$

$$\begin{aligned} P[\text{False alarm}] &= 1 - P[\text{all received are correct}] \\ &= 1 - (1 - P_e)^K \approx KP_e \\ &= 1 - (1 - 10^{-5})^{16} \\ &= 1.6 \times 10^{-4} \end{aligned}$$

False alarm often leads to retransmission

- Large K, exponential decrease in miss-detection, but linear increase in False Alarms

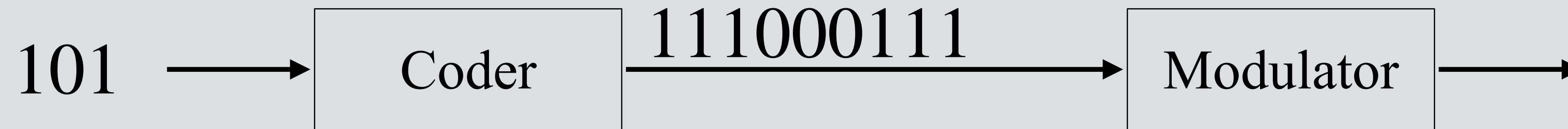


# Error Correction

- Repetition Codes - simple class, with poor performance
  - However, very low latency!
- For rate  $r < 1$  we send each bit  $1/r$  times
- $R_{\text{coded}} = rR_{\text{uncoded}}$  bps

# Repetition Codes

Consider repetition code rate  $r=1/3$ , BPSK. SNR=3dB:



$$P_e = Q(\sqrt{2}) = 0.079 \triangleq p$$

At RX, we detect each bit (i.i.d noise) and decode with majority vote

$$101 \rightarrow 1, \quad 001 \rightarrow 0$$



# Repetition Code

Bit error probability: All bits wrong and 2/3 are wrong

$$P_e[MV] = p^3 + \binom{3}{1} p^2(1-p) \\ = 0.00049 + 0.0171 = 0.0176$$

Without coding  $P_e = 0.79$ , so coding is 5 times better.

In terms of SNR:  $P_e[MV] = 0.176 = Q(\sqrt{4.44}) = 6.47\text{dB} > 3\text{dB}$

But... Energy per bit is tripled!

# Repetition Code

In terms of SNR:  $P_e[MV] = 0.176 = Q(\sqrt{4.44}) = 6.47\text{dB} > 3\text{dB}$

But... Energy per bit is tripled!

Coding gain:

$$\Delta\text{SNR} = 4.4/2 = 3.47\text{dB}$$

$$\gamma_f \triangleq r \cdot \Delta\text{SNR}$$

$$= 1/3 \cdot 4.4/2 = 0.74 = -1.3\text{dB}$$

Coding LOSS, not gain 😭

If we would transmit BPSK with  $3 \times E_b$ :

$$P_e = Q(\sqrt{3\text{SNR}}) = Q(\sqrt{6}) = 0.0072$$

# Next Time...

- Convolutional codes
- Viterbi decoding
- Block Codes